Application Web Flow Document

AI-Powered Construction Site Intelligence Platform

Version 1.1 | Status: Engineering-Ready

1. User Journey Overview

The platform has three distinct user journeys based on role. Each journey is described below from the moment the user opens the application to the completion of their primary task.

1.1 Worker Journey

- Open app → Login screen → Enter credentials → Redirect to Hazard Report submission form (workers have no dashboard access)
- Fill in hazard description, upload image, confirm location → Submit → AI categorizes hazard → Confirmation screen shows category assigned
- Worker passes through physical gate → Camera auto-verifies PPE → No app action required for entry check

1.2 Supervisor Journey

- Login → Dashboard (site-scoped view) → See live KPI cards: compliance rate, active violations, today's delivery count
- Receive real-time alert toast (new violation) → Click alert → Violation detail screen → View snapshot + metadata → Acknowledge
- Navigate to Deliveries → Log Delivery form → Enter: manual vehicle count, material type (cement/sand/bricks/steel/pipes/other), quantity, unit → Submit → System auto-compares with AI vehicle count
- Navigate to Hazard Reports → Review submitted reports with AI-assigned categories → Escalate if needed

1.3 Administrator Journey

- Login → Dashboard (all-site aggregate view) → Review escalated discrepancy queue prominently on screen
- Click discrepancy → Side-by-side: AI vehicle count vs. manual reported count, delta highlighted, supervisor who filed shown, material logged shown → Mark as resolved or flag for investigation
- Navigate to Users → Add/edit/deactivate users (workers, supervisors) → Assign to site/gate
- Navigate to Zones → Draw or define polygon restricted zone → Set zone type (crane, high voltage, excavation) → Save
- Navigate to Reports → Export compliance/material inflow reports

2. Navigation Flow

Navigation is sidebar-based. Sidebar items are role-filtered — workers only see hazard report submission; supervisors see all operational screens; admins see all screens plus user management and escalations.

Sidebar Item	Admin	Supervisor	Worker
Dashboard	Full (all sites)	Site-scoped	No access
Live Cameras	Yes	Yes (own site)	No access
Violations	Yes	Yes (own site)	No access
Zones Management	Yes (edit)	View only	No access
Equipment	Yes	Yes	No access

Sidebar Item	Admin	Supervisor	Worker
Deliveries & Materials	Full + escalations	Log + view own	No access
Hazard Reports	View all	View + manage	Submit only
User Management	Full	No access	No access
Settings	Full	Limited	No access

3. Screen-to-Screen Flow

3.1 Violations Flow

- Dashboard → (click alert card or sidebar) → Violations List → (click row) → Violation Detail → Acknowledge button → Status updates to Acknowledged → Back to list

3.2 Delivery Logging Flow

- Dashboard → Deliveries (sidebar) → Delivery Log List → "Log New Delivery" button → Log Delivery Form
- Form fields: Gate (dropdown), Manual Vehicle Count (number), Material Type (dropdown: cement/sand/bricks/steel/pipes/other), Quantity (decimal), Unit (dropdown: bags/tons/cubic_meters/pieces) → Submit
- System writes to delivery_logs, triggers AI count comparison → If mismatch pattern detected → Discrepancy flagged → Admin notified (supervisor NOT shown escalation)

3.3 Discrepancy Escalation Flow (Admin only)

- Admin Dashboard → Escalation Queue card (highlighted in amber/red) → Discrepancy List → Click row → Discrepancy Detail
- Detail shows: AI count, Manual count, Delta, Supervisor who filed, Gate, Timestamp, Material logged (informational only, not part of fraud logic) → Admin actions: Resolve / Flag for investigation / Request supervisor explanation

3.4 Hazard Report Flow

- Worker login → Hazard Report form → Upload image → Enter description + location → Submit → AI categorizes (fire risk / unsafe wiring / water leakage / scaffold damage / structural cracks) → Confirmation
- Supervisor login → Hazard Reports (sidebar) → See list with AI categories → Click → Detail view → Acknowledge / Escalate to Admin

3.5 Material Inflow Log Flow

- Supervisor/Admin → Deliveries → Material Inflow tab → Table: Date, Gate, Material Type, Quantity, Unit, Logged By → Filterable by material type and date range
- Admin can export as CSV for site reporting

4. Authentication Flow

4.1 Login

- User opens app → Redirected to /login if no valid token in storage
- Enter email + password → POST /auth/login → Server validates credentials, hashes password check
- Success: receive access_token (15 min) + refresh_token (7 days) → Store tokens → Redirect to role-appropriate landing page
- Admin → Dashboard | Supervisor → Dashboard | Worker → Hazard Report form
- Failure: show "Invalid credentials" — do not reveal whether email or password was wrong specifically

4.2 Token Refresh

- API call returns 401 (expired access token) → Frontend intercepts → Automatically POST /auth/refresh with refresh token
- Success: new access token returned → Retry original request transparently → User never sees interruption
- Refresh token also expired → Clear tokens → Redirect to login with "Session expired, please log in again"

4.3 Logout

- User clicks logout → Clear access + refresh tokens from storage → Redirect to /login

5. Admin Flow (detailed)

- Login → Full dashboard: all-site KPI aggregation
- Escalation queue visible on dashboard if unresolved discrepancies exist
- User management: create user (assign role + site), deactivate user, reset password
- Zone management: create polygon zone (draw on map or enter coordinates), assign zone_type, configure PPE rules per zone
- Reports: select date range + module (violations / deliveries / materials / hazards) → Export PDF/CSV
- Alert configuration: set escalation threshold (how many mismatches before escalation), notification method (email/in-app)

6. Role-based Access Flow

Action	Admin	Supervisor	Worker
View dashboard KPIs	All sites	Own site only	No
Acknowledge violation	Yes	Yes (own site)	No
Log delivery count + material	Yes	Yes	No
View discrepancy escalations	Yes	No (conflict of interest)	No
Submit hazard report	No	No	Yes
View hazard reports	Yes	Yes	No
Create/edit zones	Yes	No	No
Create users	Yes	No	No
Export reports	Yes	Limited	No
View material inflow log	Yes	Yes (own site)	No

7. Edge Cases

Edge Case	Expected Behaviour
Multiple workers enter gate simultaneously (tailgating)	System detects >1 person in frame during single gate cycle — flags as "multiple entry unverified" — supervisor alerted to manually verify
Camera feed drops mid-shift	Dashboard shows "Camera Offline" warning on affected gate/zone — does not silently show "no violations" — time window marked as unmonitored in audit log
Supervisor submits delivery count before AI count is logged	System holds discrepancy comparison until AI count arrives — does not compare against null — delivery log status shown as "Pending AI Count"
Worker submits hazard report with unrecognized category	AI assigns best-match category with confidence score — if below threshold, routes to "Uncategorized" queue for supervisor manual review
Expired token mid-session	Frontend silently refreshes token and retries request — user sees no interruption unless refresh token also expired, in which case graceful redirect to login
Admin tries to access own discrepancy (rare edge case)	System still routes escalation to Admin role — no self-exclusion logic required since Admin is not the filing Supervisor

Edge Case	Expected Behaviour
Duplicate delivery log for same gate same timestamp	Unique constraint on gate_id + logged_at (rounded to minute) — returns 409 Conflict with clear message
Invalid material type entered	Backend rejects with 422 — allowed list: cement, sand, bricks, steel, pipes, other — frontend dropdown prevents this in practice

8. Flow Diagrams

8.1 Main Application Flow

[Login] → { Role? }

Admin → [All-Site Dashboard] → [Escalation Queue] / [User Mgmt] / [Zones] / [Reports]

Supervisor → [Site Dashboard] → [Violations] / [Deliveries+Materials] / [Hazard Reports]

Worker → [Hazard Report Form]

8.2 Gate Detection Pipeline Flow

[Gate Camera] → [YOLOv8 Inference]

PPE branch: { Helmet + Vest detected? }

Yes → [Log Compliant Entry] → [Dashboard count update]

No → [Log Violation] → [Screenshot saved] → [Alert to Supervisor]

Vehicle branch: [Auto-increment AI vehicle count] → [Timestamped log entry]

↓ (when Supervisor later logs manual count + material)

{ Repeated count mismatch? }

Yes → [Flag discrepancy] → [Escalate to Admin only]

No → [Close — no action]

8.3 Delivery Logging Flow

[Supervisor: Log Delivery Form]

Enter: vehicle count + material type + quantity + unit

→ POST /deliveries

→ Backend: write delivery_logs record

→ Check: AI count available?

No → Status: "Pending AI Count"

Yes → Compare counts → Pattern check → Flag or close

8.4 Authentication Flow

[/login] → POST /auth/login → { Valid? }

Yes → Issue access_token (15min) + refresh_token (7d) → Redirect by role

No → "Invalid credentials" — no specifics revealed

[API call returns 401] → Auto: POST /auth/refresh

Success → Retry request

Fail → Clear tokens → /login